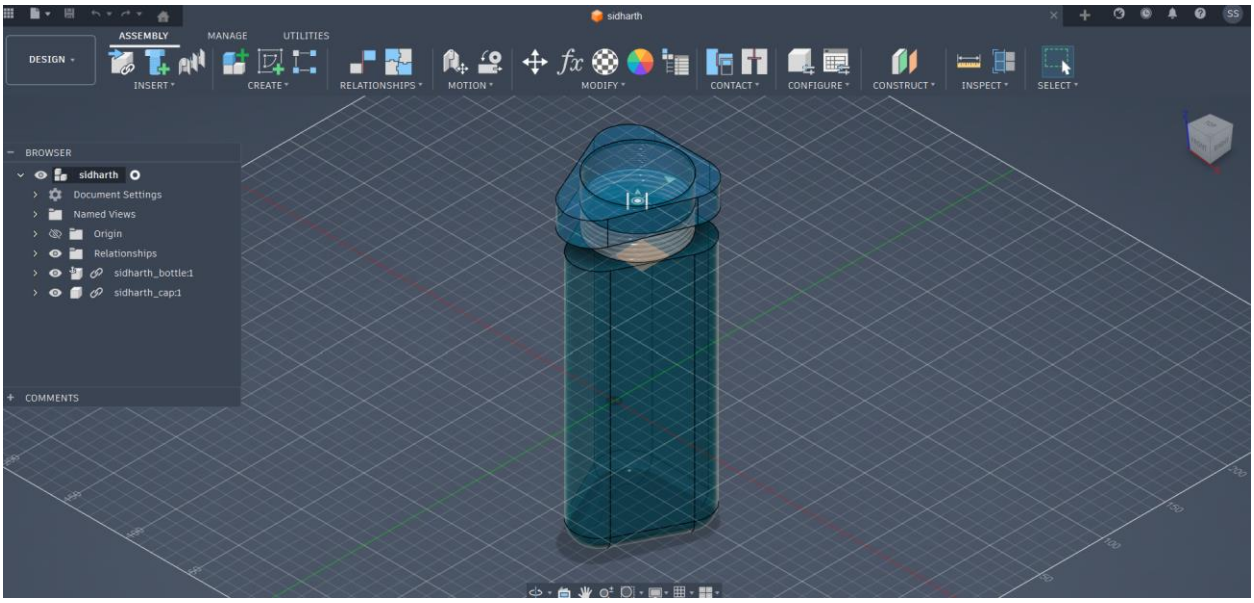
Bottle Design Assignment – Unique Water Bottle (Fusion 360)



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1. Introduction

This project is a **3D bottle design created in Autodesk Fusion 360**.
The model consists of two main components:

- **Bottle Body** (`sidharth_bottle:1`)
- **Bottle Cap** (`sidharth_cap:1`)

The design is assembled properly and shown in transparent mode to clearly visualize internal features like **threads and cap fitting**.

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2. Special / Unique Features of My Bottle Design

□ 1. Curved Ergonomic Body

- The bottle has a **smooth cylindrical structure**.
- The bottom is slightly rounded for better strength and aesthetics.

- The smooth surface makes it comfortable to hold.

□ 2. Threaded Neck Design

- The neck of the bottle contains **internal threads**.
- The cap has **matching external threads**.
- This ensures:
    - Tight sealing
    - Leak-proof functionality
    - Proper mechanical locking

□ 3. Wide Cap Design

- The cap is designed wider than the neck.
- It gives:
    - Better grip while opening
    - Modern stylish appearance
    - Stronger sealing surface

□ 4. Transparent Visualization

- The model is shown in **transparent mode**.
- This highlights:
    - Internal threading
    - Wall thickness
    - Assembly alignment

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3. Tools Used in Fusion 360

In this assignment, the following tools were used:

- **Sketch** – To draw base circle and profile
- **Extrude** – To create bottle body
- **Shell** – To make it hollow
- **Thread Tool** – To create bottle and cap threads
- **Fillet** – To smooth edges
- **Assembly / Joint** – To assemble bottle and cap
- **Appearance Tool** – To apply transparent material

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4. Design Process (Step-by-Step Explanation)

1. Created a **base circle sketch** for bottle diameter.
2. Used **Extrude** to give height.
3. Applied **Shell** command to create hollow interior.
4. Designed the **neck section**.
5. Applied **thread feature** on neck.
6. Created a separate **cap component**.
7. Applied matching internal thread to cap.
8. Assembled using **Joint tool**.
9. Applied transparent appearance for visualization.

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5. Why My Design is Unique

✓ Clean cylindrical structure
✓ Proper mechanical thread modeling
✓ Realistic assembly
✓ Transparent engineering view
✓ Balanced aesthetic + functionality

This design is not just a simple bottle — it is a **functional engineering model** suitable for manufacturing.

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6. Conclusion

This bottle assignment demonstrates:

- 3D modeling skills
- Assembly design
- Thread creation
- Engineering understanding of product design

It shows that I can create **realistic industrial product models** using Fusion 360.